



Walking Into Money: How Walk Rate Influences MLB Salary

Teddy Chisholm

Research Problem & Motivation

Main Question:

Does walk rate really affect how much players get paid in Major League Baseball?



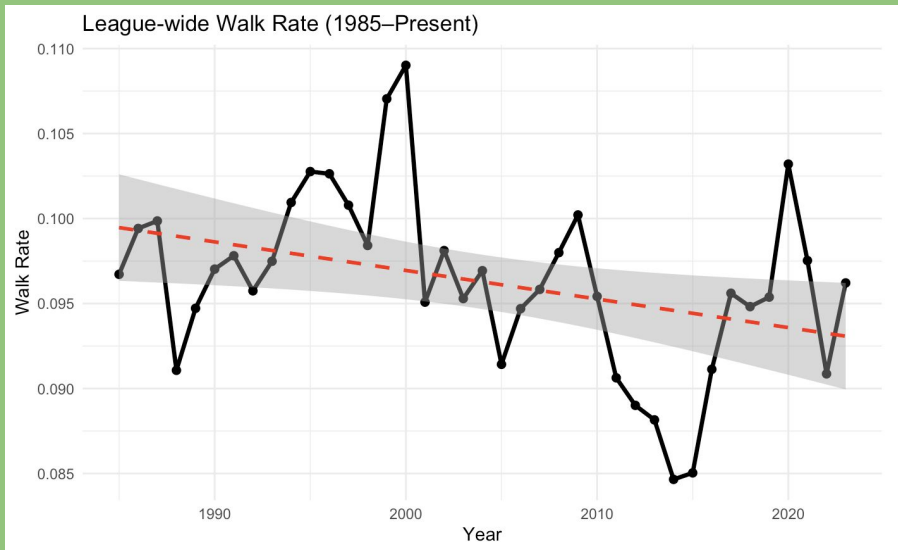
Definition:

A base on balls (BB), otherwise known as a walk is when a hitter takes 4 balls outside the strike zone before another result occurs.

Why it matters:

- Walks increase on base percentage (OBP) & run production.
- Pitchers who walk fewer hitters are more efficient.
- Walks are under-discussed in salary negotiations.

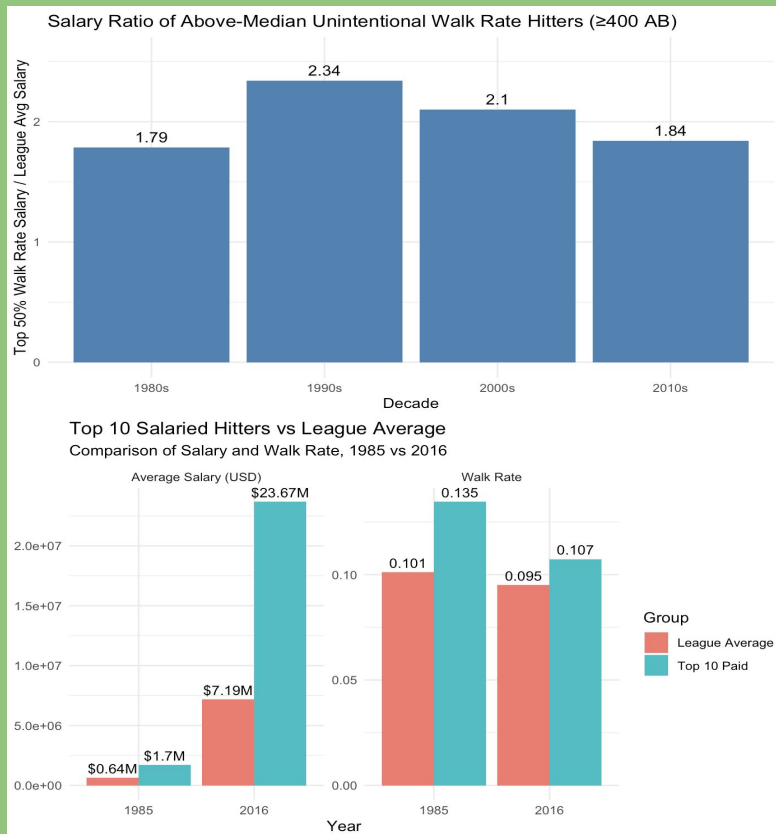
Historical Walk Trends (Hitters)



All data used from Lahman Baseball Database

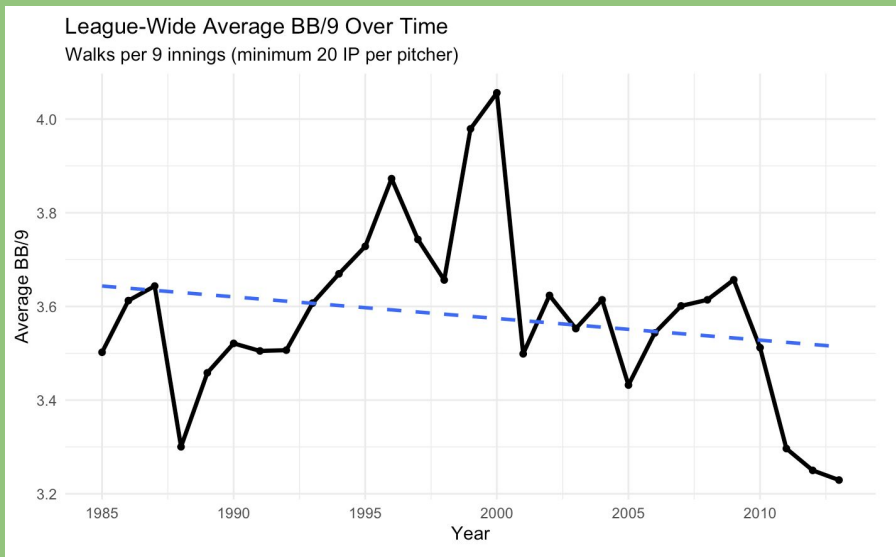
- Overall decline since 1985
- Spike in 1990s and 2000s due to the “steroid era”.
 - A large amount of players were suspected of taking PEDs in the late 90s and early 2000s.
 - Pitchers did not want to give up home runs to these guys so they walked them more often.

Hitter Salary Comparison



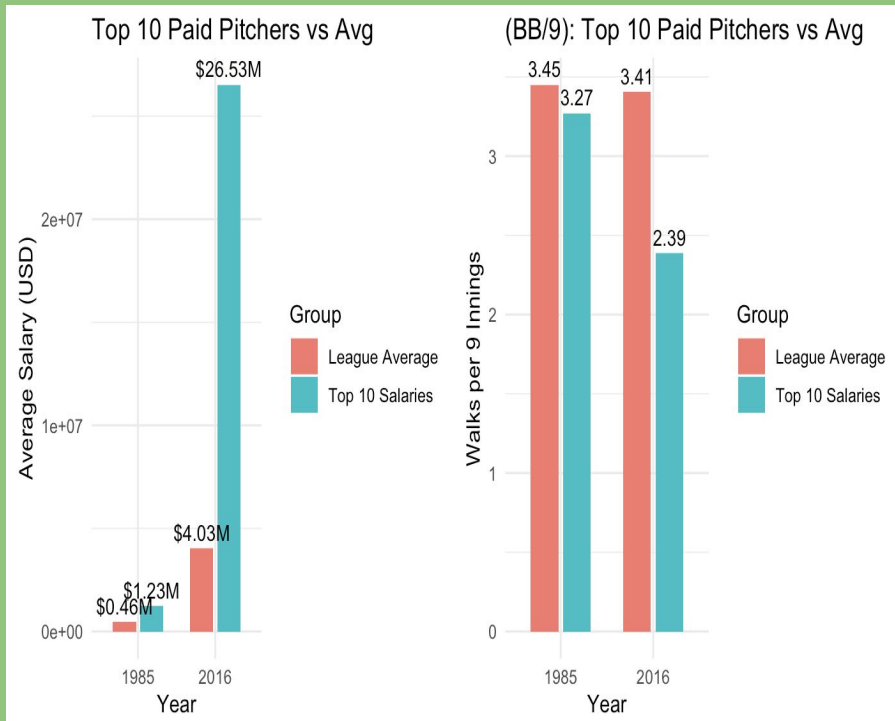
- High-walk hitters consistently earn above league-average salaries.
- Top 10 paid hitters had higher walk rates than league average (1985 & 2016).
- The highest paid hitters in 1985 walked more than in 2016.

Pitchers & Walk Trends



- **BB/9 is on the decline since 1985.**
 - Again the spike in the late '90s and early '00s is from the steroid era.
- **Most likely due to jumps in velocity, scouting data, and bullpen specialization.**

Pitcher Salary Comparison



- **1985:** The highest paid pitchers only walked slightly less batters than the league average pitcher.
- **2016:** The highest paid pitchers walked significantly less batters than the average pitcher.
- The average walk rates in 1985 and 2016 are similar.

Correlation Summary & Interpretation

- **Hitters:** Higher walk rate → higher salary
- **Pitchers:** Lower BB/9 → much higher salary

Why?

- Walks influence OBP and OPS greatly, which are driving factors in negotiations (especially OPS).
- Pitchers who walk less people allow less baserunners and therefore gives them less opportunities to allow runs.

Conclusion & Limitations

Conclusion:

- Despite not being the most looked at statistic, players who walk more get paid
- Walk rate has clear correlations with salary trends
- Especially strong for pitchers

Limitations:

- Salary data available only through 2016
 - Due to privacy policies with the MLBPA
- Correlation $\neq$ causation
- Confounding variables not discussed

Thank You!